

Revision Notes for Class 12 History Chapter 12 Security Aspects

Safety and security are major concerns when using computers and networks in today's highly connected world. While a computer with no external connection is least likely to get compromised, this is not practical for most users. Computer security involves preventing unauthorized access to data and devices, and being aware of multiple possible threats.

Connections to external devices or the Internet expose computers to various risks. Threats include hacking, malware infection, unauthorized data use, or damage to files and reputation. Effective network security consists of using technology, regular updates, and user awareness to reduce these risks.

Malware and Its Types Malware stands for “malicious software”—software designed to harm computers or data. It can steal, delete, or corrupt user data, slow down systems, or allow outside control of devices. There are several types of malware, all with their unique purposes and methods of spreading.

- **Viruses:** Code that attaches itself to legitimate programs. It only spreads when infected files are run. Viruses can corrupt files, spam contacts, or steal information. Examples include CryptoLocker and ILOVEYOU.
- **Worms:** Standalone programs that self-replicate across networks, causing widespread damage without human help.
- **Ransomware:** Locks or encrypts your files and demands money (ransom) to unlock them. A famous case is WannaCry, which affected thousands of computers in 2017.
- **Trojans:** Disguised as harmless software but give attackers control of your system when installed.
- **Spyware:** Secretly collects user data and sends it to external sources, often without notice.

- **Adware:** Bombards users with unwanted advertisements and may expose them to more serious malware via unsafe links.
- **Keyloggers:** Record keystrokes and threaten sensitive information like passwords.

Ways That Malware Spreads

Malware is distributed by several methods, putting even careful users at risk. Internet downloads—especially of “free” programs—not checked by antivirus can install malicious code.

Email attachments or suspicious links in spam messages are very common pathways. Sharing removable devices like USB drives without scanning for threats can transfer malware. Some types, like worms, spread automatically over connected networks with no action needed by the user.

Signs of Malware Infection

A malware-infected computer can show clear warning signs. For example, too many popups, changed browser settings, or slow and crashing systems.

Sending spam emails from your own account, unexpected programs running during startup, files missing or appearing on their own, and storage filling up quickly also indicate problems. If you notice any of these, suspect malware and take immediate action.

Malware Prevention Tips

Good habits and security tools make a big difference in keeping your systems safe. Use and update trusted antivirus software. Keep your operating system and applications updated to fix security holes. Review your browser security settings and look for the “https://” sign on websites, especially when making payments.

Avoid pirated programs and suspicious downloads. Back up important data regularly and use firewalls to restrict unwanted access. Be cautious with removable drives and never share login credentials or install unverified security tools, especially those offered in popups.

Uninstall unknown or suspicious programs and always close unwanted popups from the corner "X," not from within the popup. Cancel installations you did not start. Each of these steps helps prevent infection and keeps your information protected.

Antivirus and How It Works

Antivirus software, now often called anti-malware, doesn't just remove viruses but scans for all types of bad software. It protects your device in various ways.

- **Signature-based detection:** Compares files to a database of known threats. Needs updates to stay effective.
- **Sandbox detection:** Runs suspicious files in a separate, safe environment to see if they act like malware.
- **Data mining:** Uses machine learning to classify what is safe or suspicious.
- **Heuristic analysis:** Looks for code patterns that resemble known threats, even if a virus is new.
- **Real-time protection:** Watches what your applications are doing all the time.

Spam and Its Risks

Spam refers mainly to unwanted bulk messages, often emails filled with ads or scams. These could fill your inbox or even install malware with a single click if you're not careful.

Email services such as Gmail and Hotmail have inbuilt tools to separate spam from useful messages. You can mark suspicious mail as spam yourself to further protect your account.

Understanding HTTP vs HTTPS

HTTP is the standard protocol for loading web pages, but it is not secure—data is sent in plain text and can be intercepted by someone on the same network.

HTTPS encrypts data so your communications (such as passwords or bank information) are concealed from others. Always use only “https://” sites for any financial or personal transactions and look for a lock symbol near the website address.

Firewalls and Their Importance

A firewall is a barrier that helps block unauthorized traffic coming into or out of your computer or network. It acts as a gatekeeper between trusted and untrusted networks. There are network firewalls (protecting whole networks) and host-based firewalls (protecting individual devices). They monitor and control who can connect, helping keep attackers away.

Cookies—Convenience with Caution

Cookies are small data files a website stores on your computer. They help remember what’s in your shopping cart or keep you logged in as you move from page to page. Common types of cookies are session cookies (end when you log out), authentication cookies (keep you logged in), and those that autofill forms for you. But cookies can be exploited: “supercookies” or “zombie cookies” can track users across sites and are hard to remove. Be mindful of giving permission and check your browser’s settings to manage or delete cookies.

Who Are Hackers and Crackers?

People with advanced computing skills can become either helpful or harmful to network security. “White hat” hackers, or ethical hackers, help organizations find and fix weaknesses.

“Black hats,” called crackers, use skills unethically to break into systems. “Grey hats” fall in between—curious or mischievous, but not always with harmful intent.

Network Security Threats

Network threats are serious risks to both individuals and organizations. Denial of Service (DoS) attacks flood a site or server to block it from genuine users, and Distributed Denial of Service (DDoS) attacks use a network of infected devices (“botnets”) to make blocking them extremely difficult.

Network intrusion involves unauthorized activities within a network, like exploiting weaknesses (buffer overflow), sneaking traffic past sensors (asymmetric routing), or overwhelming monitoring software (traffic flooding).

Snooping and eavesdropping are other common network threats. Snooping means passively capturing or analyzing network traffic, often to find sensitive data or passwords, using specialized software.

Eavesdropping is even more direct: it is real-time listening to communications (calls, chat, video) without permission, usually by hacking devices or intercepting network signals. Both endanger privacy and safety, so safeguards and regular checks are vital.

In summary, being aware of threats—from malware, spam, and insecure websites to hackers and network attacks—helps you adopt habits to stay safe online. Using updated security tools, practicing caution, and understanding how attacks work form the core of effective cyber security.